

Aidan Gardner-O'Kearny

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Education

PhD	Univeristy Of Tennessee, Knoxville, Physics	Aug 2025 - Present
MS	Univeristy Of Tennessee, Knoxville, Physics	Aug 2025 - Present
BS	University of Oregon, Physics - Clark Honors College	Sept 2021 – Jun 2025
• Honors Thesis: Combinatorial fake discriminatory selections for combined dE/dx and disappearing track search with the ATLAS detector		

Research Experience

Holmes Group - Graduate Research Assistant	Knoxville, TN
• Worked as a part of the MAIA detector group • Characterized photon reconstruction within PandoraPFA framework and optimized tagging criteria	May 2026 - Present
Jeanty Group - Undergraduate Research Assistant	Eugene, OR
• Worked with Dr. Laura Jeanty as a part of the ATLAS Collaboration • Part of upcoming dE/dx and disappearing track signature analysis for long-lived, massive, charged particles	Jun 2023 – Jun 2025
Brookhaven National Lab - SULI Intern	Upton, NY
• Worked with Dr. Abraham Tishelman-Charny as a part of the ATLAS Collaboration • Member of upcoming $SH \rightarrow b\bar{b}\gamma\gamma$ analysis searching for beyond Standard Model scalar bosons	Jun 2024 – Aug 2024

Teaching Experience

UTK Physics Department - Graduate Teaching Assistant	Knoxville, TN
• Instructor for general physics for engineers	Sept 2025 - Present
UO Tutoring and Academic Engagement Center - Class Encore Mentor	Eugene, OR
• Leadership role within the Class Encore program	Jun 2024 - Jun 2025
UO Tutoring and Academic Engagement Center - Class Encore Leader	Eugene, OR
• Support students in introductory courses in order to increase confidence in lecture material	Sept 2022 - Jun 2025
Oregon Pathways to Industry Research Careers - OPIRC Tutor	Eugene, OR
• Multivariable calculus tutor	Apr 2025 - Jun 2025
UO Physics Department - Introductory Physics Learning Assistant	Eugene, OR
• Member of the general physics educational team	Apr 2024 - Jun 2024

Presentations

Combinatorial fake discriminatory selections for combined dE/dx and disappearing track search	May 2025
• UO Undergraduate Research Symposium • Oral presentation • Poster presentation • Aidan Gardner-O'Kearny	

dE/dx and disappearing track background discrimination studies	April 2025
• UO Clark Honors College Three Minute Thesis • Oral presentation • Aidan Gardner-O’Kearny	
Background ionization behavior for combined dE/dx and disappearing track analysis in the ATLAS detector	March 2025
• APS Global Summit • Oral Presentation • Aidan Gardner-O’Kearny, Laura Jeanty, Nathan Young	
Comparing resolved and boosted jet identification algorithms to search for beyond the Standard Model scalar bosons with the ATLAS detector	August 2024
• Brookhaven SULI Intern Research Fair • Poster Presentation • Aidan Gardner-O’Kearny, Abraham Tishelman-Charny, Elizabeth Brost	
Background ionization behavior for combined dE/dx and disappearing track analysis	May 2024
• UO Undergraduate Research Symposium • Poster Presentation • Aidan Gardner-O’Kearny, Laura Jeanty, Nathan Young	

Awards

UO Physics Department Weiser Research Award	June 2025
UO Physics Department Poster Award - 2nd Place	June 2025
• Combinatorial fake discriminatory selections for combined dE/dx and disappearing track search	
UTK Graduate Fellowship	Feb 2025
UO CURE Travel Grant Awardee	Nov 2024
• Award given to support undergraduates presenting their research at a relevant conference	
UO Summit Scholar	Sept 2021
• Academic excellence scholarship	

Outreach

Eugene Science Center	Nov 2024
• Organized an outreach event for elementary school students • Provided undergraduate students opportunity to perform demos	

Membership

Phi Beta Kappa	June 2025
American Physical Society	Oct 2024 - Present
• Student Member	
Society Of Physics Students	Jun 2023 - Jun 2025
• UO Chapter Treasurer	

Skills

Languages: Python, C++, JULIA, Bash, T_EX/Markdown, Mathematica

Technologies: Linux, Windows, ROOT, Matplotlib/MPLHEP, Git, PandoraPFA, Google And Microsoft Office Suite